

Blockchain Course Content

Beginner to Intermediate | Hinglish | 8 Weeks | Self-paced or Instructor-led

Course Objective

Is course ka goal learners ko blockchain ki basic theory, practical use cases, wallet concepts, smart contracts, DApps, security practices, aur career roadmap samjhana hai. Course complete karne ke baad learner blockchain ecosystem ko clearly explain kar sakega aur basic decentralized application ka plan bana sakega.

Target Audience

Students jo blockchain technology start karna chahte hain.
Developers jo Web3 aur smart contracts seekhna chahte hain.
Business professionals jo blockchain use cases samajhna chahte hain.
Beginners jinko cryptocurrency aur decentralized systems ka overview chahiye.

Prerequisites

Basic computer knowledge.
Internet aur browser usage ka experience.
Programming knowledge helpful hai, lekin mandatory nahi.
Developer track ke liye JavaScript / TypeScript basics recommended hain.

Week 1: Blockchain Fundamentals

Topics:

- Blockchain kya hai?
- Centralized, decentralized, aur distributed systems.
- Blocks, transactions, hashes, nonce, aur chain structure.
- Immutability aur transparency.
- Public vs private blockchains.

Learning Outcomes:

- Learner key concepts simple language me explain kar sakega.
- Learner practical examples aur limitations identify kar sakega.

Activity: Ek simple block diagram banaiye: transaction data, previous hash, current hash, timestamp, aur nonce.

Week 2: Cryptography Basics

Topics:

- Hashing kya hoti hai?
- SHA-256 ka high-level overview.
- Public key aur private key ka role.
- Digital signatures.
- Wallet address kaise banta hai?

Learning Outcomes:

- Learner key concepts simple language me explain kar sakega.
- Learner practical examples aur limitations identify kar sakega.

Activity: Online hash generator me text ka hash banakar observe kariye ki small change se output completely change hota hai.

Week 3: Consensus Mechanisms

Topics:

- Consensus ki need kyun hoti hai?
- Proof of Work.

- Proof of Stake.
- Validators, miners, staking, gas fees.
- Forks aur finality.

Learning Outcomes:

- Learner key concepts simple language me explain kar sakega.
- Learner practical examples aur limitations identify kar sakega.

Activity: Proof of Work aur Proof of Stake ke pros and cons ki comparison table banaiye.

Week 4: Cryptocurrency and Wallets

Topics:

- Cryptocurrency kya hai?
- Coins vs tokens.
- Hot wallet vs cold wallet.
- Seed phrase aur backup practices.
- Exchanges, custody, aur self-custody.

Learning Outcomes:

- Learner key concepts simple language me explain kar sakega.
- Learner practical examples aur limitations identify kar sakega.

Activity: Ek testnet wallet setup ka written walkthrough banaiye. Real funds use na karein.

Week 5: Smart Contracts

Topics:

- Smart contract kya hota hai?
- Ethereum Virtual Machine ka overview.
- Solidity introduction.
- Contract deployment lifecycle.
- Events, functions, state variables, aur gas.

Learning Outcomes:

- Learner key concepts simple language me explain kar sakega.
- Learner practical examples aur limitations identify kar sakega.

Activity: Simple voting system ka paper design banaiye: voters, candidates, vote count, aur restrictions.

Week 6: Decentralized Applications

Topics:

- DApp kya hota hai?
- Frontend, wallet, smart contract, aur blockchain node interaction.
- Web3 libraries ka overview.
- Oracles ka role.
- DApp UX challenges.

Learning Outcomes:

- Learner key concepts simple language me explain kar sakega.
- Learner practical examples aur limitations identify kar sakega.

Activity: Ek DApp idea choose karke architecture draw kariye: frontend, wallet, contract, data source.

Week 7: Blockchain Use Cases

Topics:

- Finance and payments.
- Supply chain tracking.
- Identity management.

- NFTs and digital ownership.
- Healthcare records.
- Governance and voting.

Learning Outcomes:

- Learner key concepts simple language me explain kar sakega.
- Learner practical examples aur limitations identify kar sakega.

Activity: Ek use case select karke likhiye: blockchain kyun useful hai, stakeholders kaun hain, aur risks kya hain?

Week 8: Security, Legal Awareness, and Career Roadmap

Topics:

- Common scams: phishing, fake airdrops, rug pulls.
- Smart contract vulnerabilities.
- Audit ka importance.
- Regulatory awareness.
- Career paths: developer, auditor, product manager, analyst, community manager.
- Learning roadmap and portfolio ideas.

Learning Outcomes:

- Learner key concepts simple language me explain kar sakega.
- Learner practical examples aur limitations identify kar sakega.

Activity: Apna 30-day blockchain learning plan banaiye jisme reading, practice, project, aur portfolio tasks hon.

Capstone Project Ideas

1. Simple token concept document.
2. Decentralized voting system design.
3. Supply chain tracking workflow.
4. NFT certificate issuing platform plan.
5. Crypto wallet security checklist website.

Assessment Plan

Weekly quiz: 20%

Practical activities: 30%

Final capstone project: 40%

Participation / discussion: 10%

Recommended Tools

Browser-based blockchain explorers.

Testnet wallet.

Remix IDE for Solidity practice.

GitHub for portfolio.

Documentation from major blockchain platforms.

Safety Note

Learning ke liye testnets ka use karein. Real private keys, seed phrases, ya real funds kisi demo ya practice activity me share na karein.